

Roman bridges

To create structures strong enough to span large distances, while bearing heavy loads and remaining stable, the Romans constructed voussoir arches with keystones. Voussoirs were wedge-shaped or tapered blocks of stone placed in semi-circular arrangements; to each side, the arch rested on piers or columns of stone blocks mortared with pozzolanic cement. The weight of the stone and concrete in the bridge itself compressed the tapered stones together, making the arch extremely strong. Voussoir arches were the basis for both single-span bridges and lengthy multi-arched aqueducts. Some early Roman arch bridges, influenced by the ancient notion of the ideal form of the circle, are made from a full circle, with the stone arch continuing underground. However, once engineers realised that an arch did not even have to be semicircular to remain stable, some later structures featured 'segmental' arches encompassing only a smaller arc of the circle. The Romans were also the first – and, until the Industrial Revolution, only – engineers to construct concrete bridges.



Roman bridge at Giornico, Switzerland